KULLIYYAH OF INFORMATION AND COMMUNICATION TECHNOLOGY

CSCI 2304 – INTELLIGENT SYSTEM

SEMESTER 2, 2022/23

GROUP PROJECT

NAME	MATRIC NO
SHAFIA HANIS BT SHAMSUL ASMADI	2110578
BATRISYIA KARMILA BINTI AINOL AHMAM	2113836
AUNI FATINI BT SANIZA	2113244
NUR ALYA SHAZWANI BT MOHD YAZI	2112766

LECTURER

ASSOC. PROF. TS. DR. AMELIA RITAHANI

TABLE OF CONTENTS

1.0 ABSTRACT

The main goal of this research project is to create a weather prediction model utilizing historical weather information. From daily planning to strategic decision-making in sectors like agriculture, transportation, and tourism, accurate weather forecasting is essential in many parts of our lives. We seek to improve the precision and dependability of weather predictions by utilizing machine learning techniques and studying the "Seattle Weather" dataset from 2012 to 2015. Insights into Seattle's weather patterns and temperature changes are discovered through data investigation and visualization, laying the groundwork for model development. The effective application of a trustworthy weather prediction model can significantly influence decision-making procedures and resource allocation, ultimately benefiting people, companies, and society at large.

2.0 INTRODUCTION

2.0 Background

A vital field of study that significantly impacts numerous aspects of our life is weather forecasting. We can plan activities, make appropriate choices, and reduce risks linked with bad weather thanks to accurate forecasts. Accurate weather forecasting has wide-ranging effects on various sectors of the economy, including agriculture, transportation, energy, and disaster relief.

Historically, manual observations and empirical standards have been the foundation of weather forecasting. However, there has been a paradigm change in favor of data-driven techniques with the development of technology and the accessibility of large-scale weather information. By examining previous weather patterns and finding pertinent patterns and trends, machine learning and statistical models have demonstrated promise in improving the accuracy and reliability of weather predictions.

A reliable source for creating accurate weather prediction models is historical weather data. Researchers and meteorologists can recognise recurring patterns, comprehend long-term trends, and establish correlations between various meteorological variables by studying past weather conditions. This data serves as the foundation for creating prediction models that extrapolate past weather data to anticipate future weather conditions.

Research on weather prediction is becoming more accessible thanks to the availability of comprehensive meteorological datasets, such as the "Seattle Weather" dataset used in this project. The temperature, precipitation, wind speed, and weather conditions are only a few of the meteorological factors that are included in detail in these databases. Such datasets can be utilized in conjunction with machine learning algorithms to create complex models that can produce accurate and timely weather forecasts.

The creation of a trustworthy weather forecast model has broad implications for daily life. It helps transportation companies plan routes and account for weather-related disruptions, aids energy companies in managing power generation and distribution, and helps emergency management organizations get ready for severe weather events. Farmers can use it to optimize crop planting and irrigation schedules.

In conclusion, this project's history is rooted in the growing significance of accurate weather forecasting and the possibility of data-driven methodologies to raise forecasting accuracy. Our goal is to advance weather prediction techniques by combining historical meteorological data with machine learning algorithms to create a solid model that will help a variety of sectors and increase society adaptability to changing weather conditions.

2.1 Objective

The objective of this program is to develop a weather prediction model using historical weather data. Here in this project, we gather six classification models and compare with each other. This project aims to leverage machine learning algorithms and data analysis techniques to enhance the accuracy and reliability of weather forecasts. There area few specific objectives of this project which is:

2.1.1 Data Exploration

We aim to explore and analyze the “Seattle Weather” dataset and gain some insights regarding weather patterns, trends and variations over the specified time period.

2.1.2 Visualization

We also plan to create some visualizations based on the dataset using plot, and graphs. This is to visualize the pattern that we obtain and analyze the relationships between different meteorological variables.

2.1.3 Model Evaluation

After gaining the results, we aim to assses th performances and the accuracy of each classification models’ results using appropriate evaluations metrics to ensure the reliability and effectiveness.

3.0 LITERATURE REVIEW

In many different facets of our existence, weather forecasting is an essential topic of study. We may limit risks connected with adverse weather by planning activities, making educated decisions, and using accurate predictions.

3.1 The Application of Statistical Models and Machine Learning to Weather Prediction

The accuracy and dependability of weather predictions have shown promise when combined with machine learning and statistical models. These models may be trained using meteorological data from the past to discover the patterns and trends that forecast weather conditions in the future.

The linear regression model is one of the most widely used machine learning techniques for forecasting the weather. Based on a linear relationship between the current and previous weather conditions, this model forecasts the future weather.

This random forest model is another popular machine learning method used for weather forecasting. A variety of decision trees are used in this ensemble model to produce predictions.

3.2 The Availability of Comprehensive Meteorological Datasets

It is now simpler to conduct research on weather prediction since extensive meteorological databases are readily available. A wide range of meteorological factors, such as temperature, precipitation, wind speed, and weather condition, are all included in these databases in great detail.

The Global Historical Climatology Network (GHCN) is one of the largest meteorological databases. Over 100,000 weather stations from all over the world contributed weather data to the GHCN collection.

3.3 The Consequences of a Reliable Weather Forecast Model

A reliable weather forecast model affects many aspects of daily life. It can aid emergency management organizations in preparing for extreme weather occurrences, energy firms in managing power generation, and transportation corporations in planning routes.

For instance, transport firms can utilize weather forecasts to select routes that steer clear of inclement weather. Weather predictions can be used by energy corporations to control electricity generation and distribution. Additionally, organizations in charge of emergency management can utilize weather prediction to plan for dangerous weather conditions like hurricanes and tornadoes.

3.4 The Challenges of Weather Forecasting

Accurate weather predicting is still difficult, despite development in machine learning and statistical modeling. The chaotic character of the environment is one of the key difficulties, it is challenging to forecast how these factors will interact in the future since the atmosphere is a complex system with many interacting variables.

The restricted availability of historical weather data is another difficulty for reliable weather forecasting. Only a little amount of historical weather information is accessible in various regions of the world. This might make it challenging to develop statistical and machine learning models that can effectively forecast weather in these regions.

3.5 Forecasting Weather in the Future

Weather forecasting has a bright future. In the future, we may anticipate seeing weather forecasts that are even more precise because of ongoing developments in machine learning and statistical modeling.

The advancement of artificial intelligence (AI) may also completely alter weather forecasting. AI may be used to create more intricate models that can account for a larger variety of variables, such as the atmosphere's chaos and the dearth of historical meteorological data.

4.0 METHODOLOGY / EXPERIMENTAL SETUP

4.1 Methodologies

4.1.1 Data Preprocessing

To ensure the quality and reliability of the data, we performed thorough preprocessing steps. This involved removing outliers, and normalizing or standardizing the features. Additionally, we conducted exploratory data analysis to gain insights into the data distribution and identify any patterns or correlations.

4.1.2 Model Selection

We experimented with several classification models to determine the most suitable approach for weather prediction. The models we evaluated include Support Vector Machines (SVM), K-Nearest Neighbors (KNN), Naive Bayes, Decision Tree Classifier (DTR), Random Forest (RF), and XGBoost. Each model has its own strengths and weaknesses, and we aimed to assess their performance in terms of accuracy.

4.1.3 Model Training and Evaluation

We split the dataset into training and testing sets, reserving a portion for evaluating the model's performance. We trained each classification model on the training set and fine-tuned their respective hyperparameters using techniques such as cross-validation. Finally, we evaluated the models on the testing set to measure their accuracy.

4.1.4 Performance Comparison

To compare the performance of the different classification models, we calculated and recorded their accuracy scores. We visualized the results using a bar plot, highlighting the accuracy achieved by each model. This enables us to identify the most accurate model for weather prediction based on our dataset.

4.2 Flow Chart

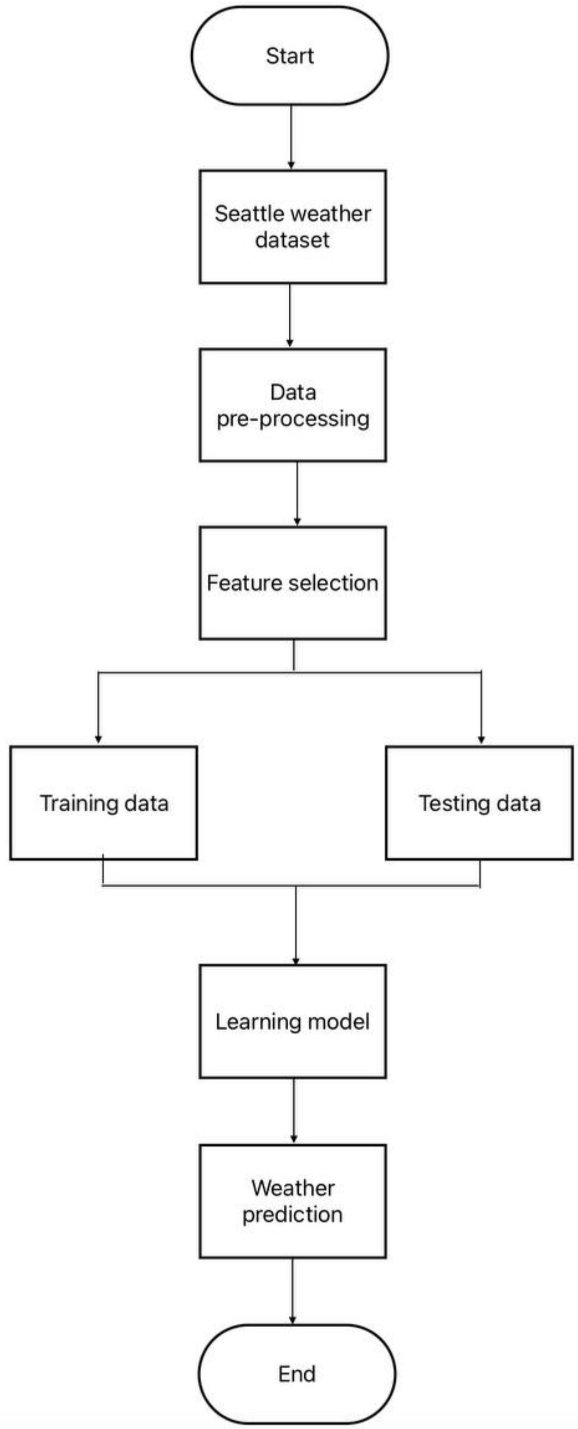

Figure 1

4.3 Pseudocode

1. Import necessary libraries:
    a. numpy
    b. pandas
    c. os
    d. matplotlib.pyplot
    e. seaborn
2. Define a function LABEL_ENCODING that takes a column name as input:
    a. Use label encoding to transform the values of the specified column
    b. Update the data with the transformed values
3. Load the dataset using pd.read_csv() and assign it to the variable 'data'
4. Preprocess the data:
    a. Convert the 'date' column to datetime format using pd.to_datetime()
    b. Perform label encoding on the 'weather' column using the LABEL_ENCODING() function
    c. Drop the 'date' column from the data
5. Split the data into features (X) and target variable (y)
6. Perform train-test split using train_test_split() from sklearn.model_selection:
    a. Split X and y into training and testing sets (X_train, X_test, y_train, y_test)
    b. Set the test size to 0.25 and random_state to a specific value
7. Standardize the features using StandardScaler() from sklearn.preprocessing:
    a. Fit and transform X_train
    b. Transform X_test
8. Train and evaluate different classification models:
    a. Logistic Regression:
        i. Create a LogisticRegression classifier
        ii. Fit the model on the training data

    iii. Predict the target variable for the test data
    iv. Calculate the accuracy score and confusion matrix
    v. Store the accuracy score in a list
b. Support Vector Machine (SVM):
    i. Create an SVC classifier with a linear kernel
    ii. Fit the model on the training data
    iii. Predict the target variable for the test data
    iv. Calculate the accuracy score and confusion matrix
    v. Store the accuracy score in a list
c. K-Nearest Neighbors (KNN):
    i. Create a KNeighborsClassifier with 5 neighbors and 'minkowski' distance metric
    ii. Fit the model on the training data
    iii. Predict the target variable for the test data
    iv. Calculate the accuracy score and confusion matrix
    v. Store the accuracy score in a list
d. Naive Bayes:
    i. Create a GaussianNB classifier
    ii. Fit the model on the training data
    iii. Predict the target variable for the test data
    iv. Calculate the accuracy score and confusion matrix
    v. Store the accuracy score in a list
e. Decision Tree Classifier:
    i. Create a DecisionTreeClassifier with 'entropy' criterion
    ii. Fit the model on the training data
    iii. Predict the target variable for the test data
    iv. Calculate the accuracy score and confusion matrix
    v. Store the accuracy score in a list
f. Random Forest Classifier:
    i. Create a RandomForestClassifier with 40 estimators
    ii. Fit the model on the training data

iii. Predict the target variable for the test data
iv. Calculate the accuracy score and confusion matrix
v. Store the accuracy score in a list

g. XGBoost Classifier:

i. Create an XGBClassifier
ii. Fit the model on the training data
iii. Predict the target variable for the test data
iv. Calculate the accuracy score and confusion matrix
v. Store the accuracy score in a list

9. Create two empty lists, mylist and mylist2
10. Append the accuracy scores and model names to mylist and mylist2 respectively for each model
11. Plot a bar chart using seaborn's barplot:
    a. Set the model names as x-axis and accuracy scores as y-axis
    b. Set the color palette, labels, and title of the plot
    c. Display the bar chart

5.0 RESULTS & DISCUSSION

The following figures display the system's output.

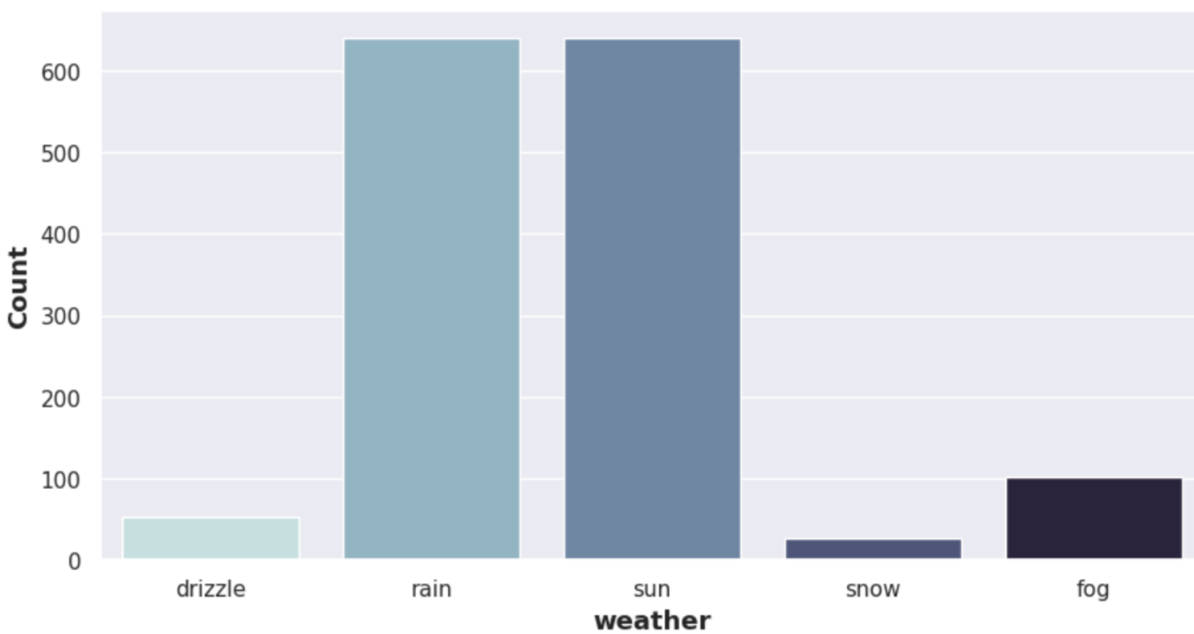

Figure 2 : Weather count

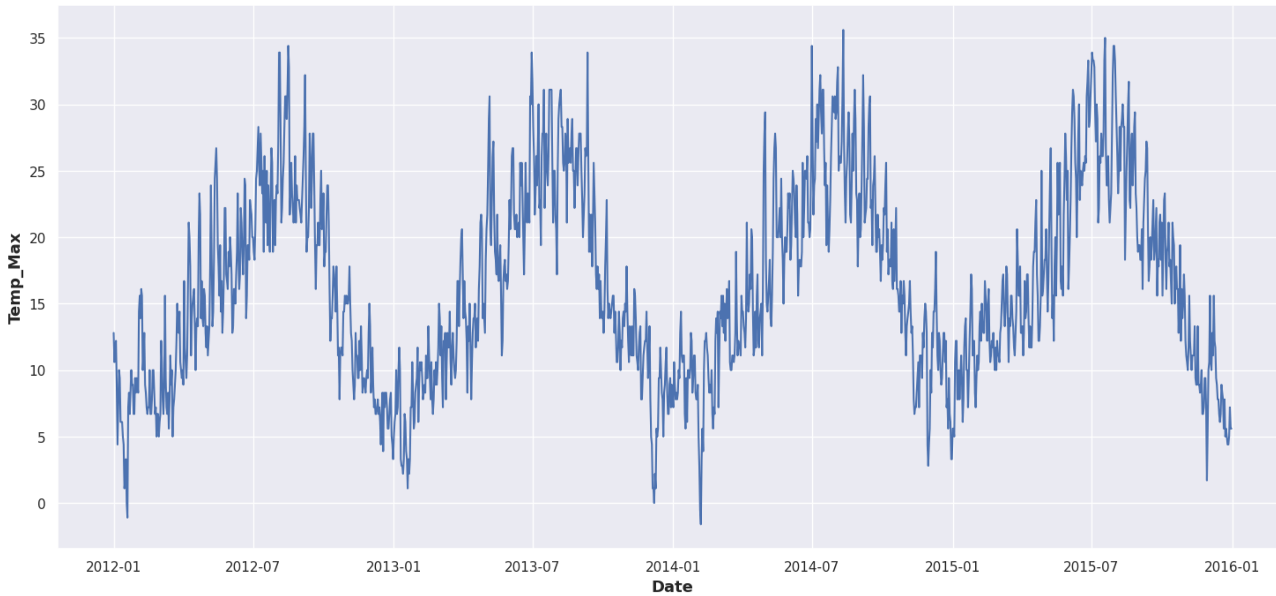

Figure 3 : Maximum temperature from Jan 2012 to Jan 2016

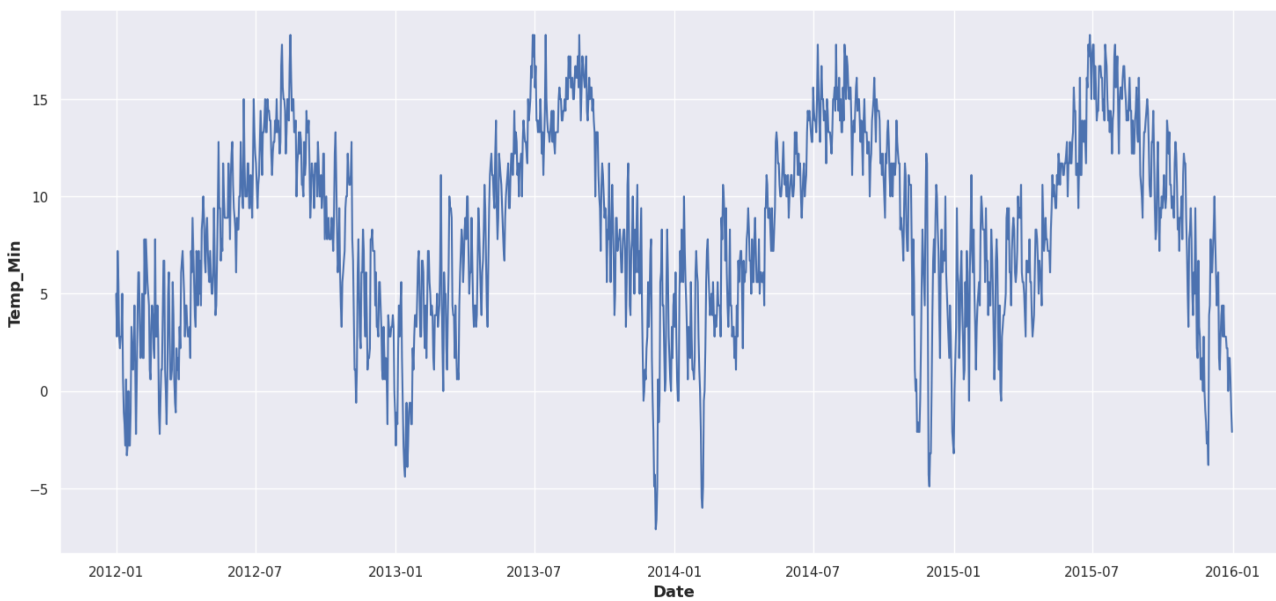

Figure 4 : Minimum temperature from Jan 2012 to Jan 2016

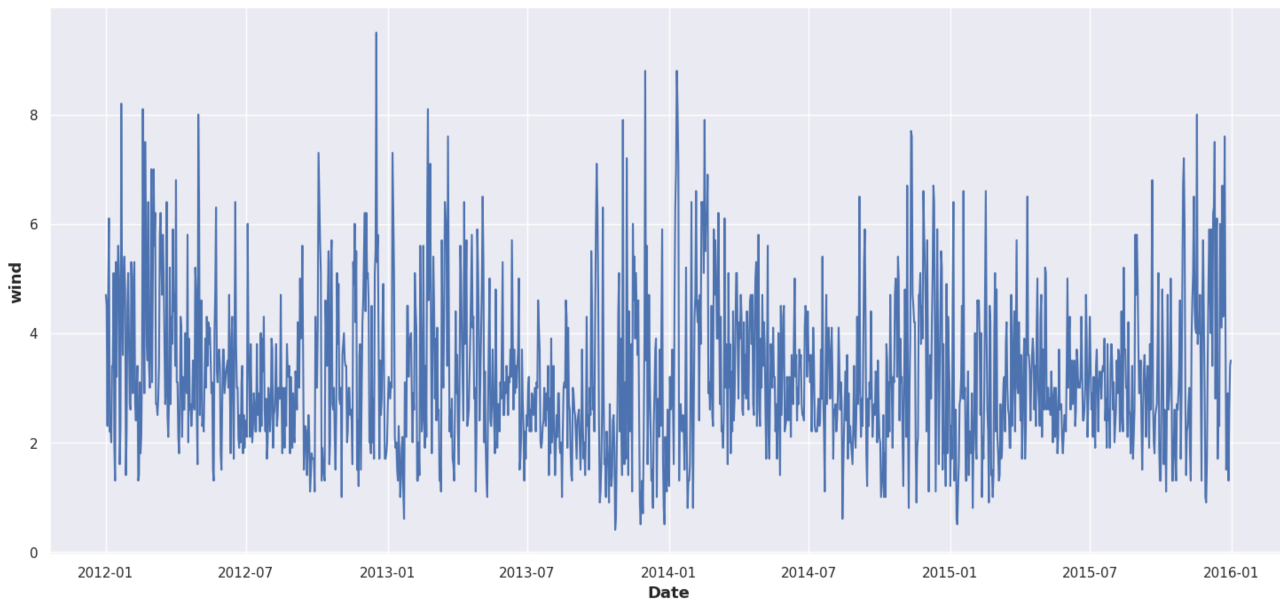

Figure 5 : Wind count from Jan 2012 to Jan 2016

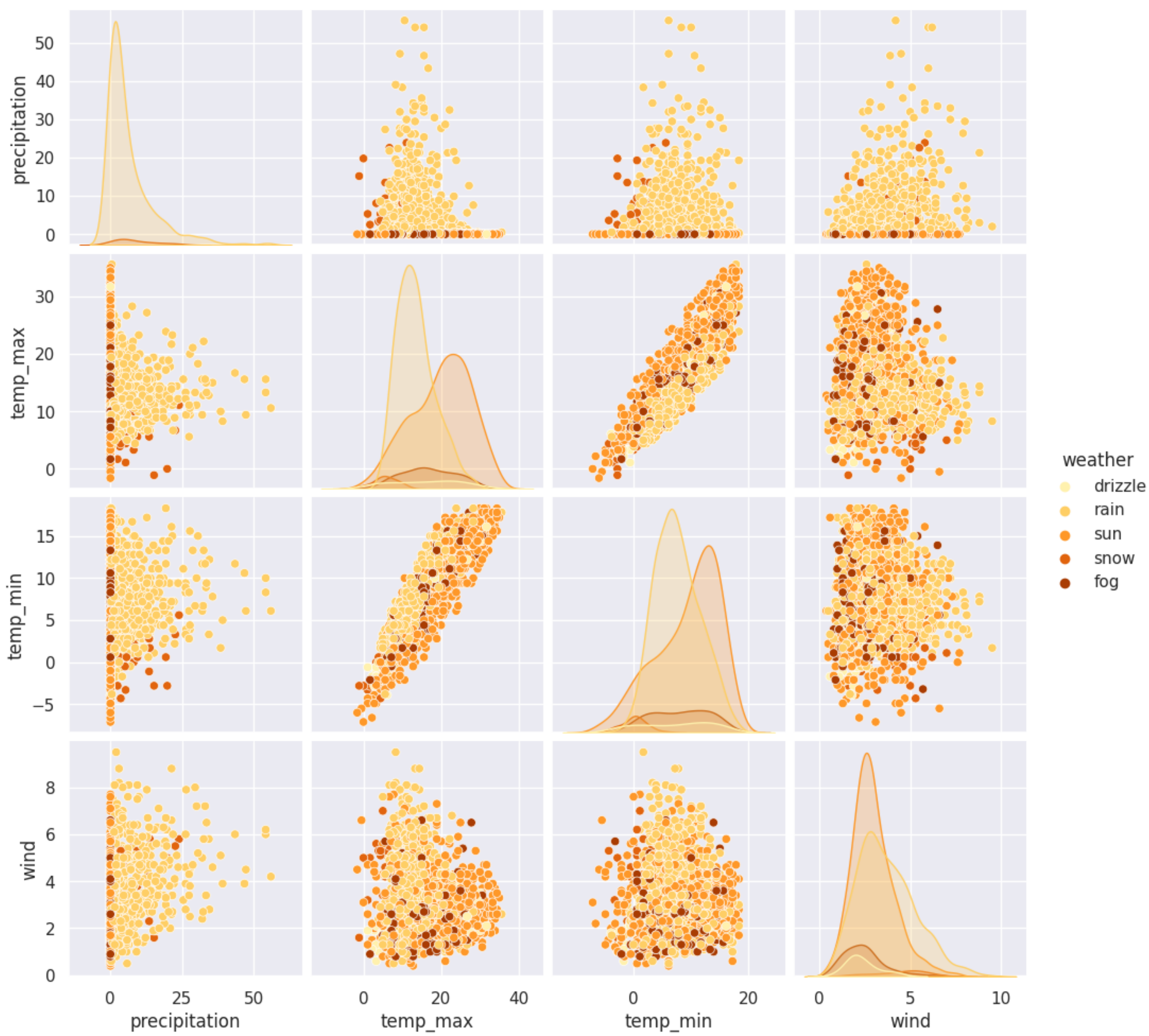

Figure 6 : Pair plot for each weather

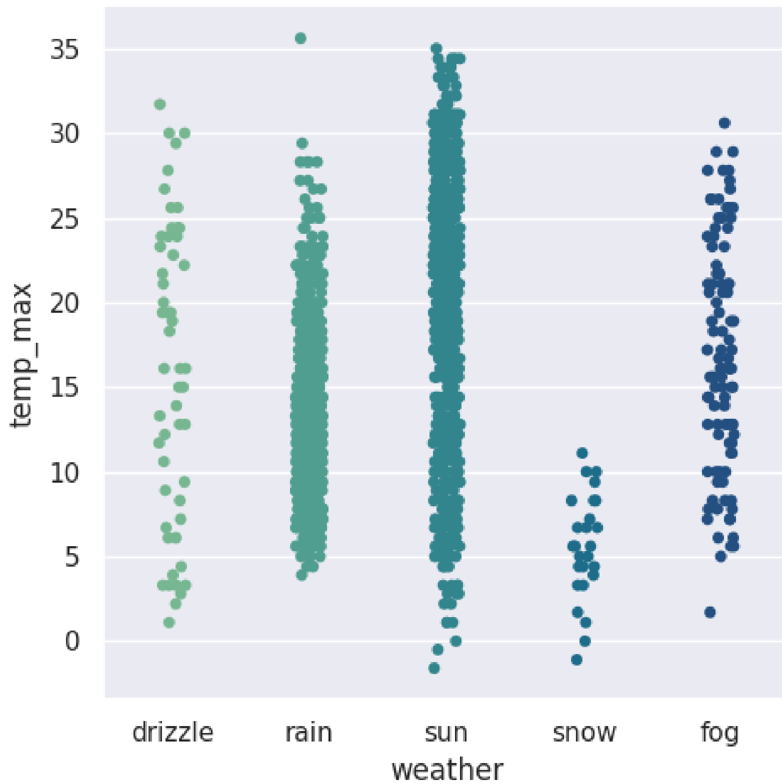

Figure 7 : Catplot for weather and maximum temperature

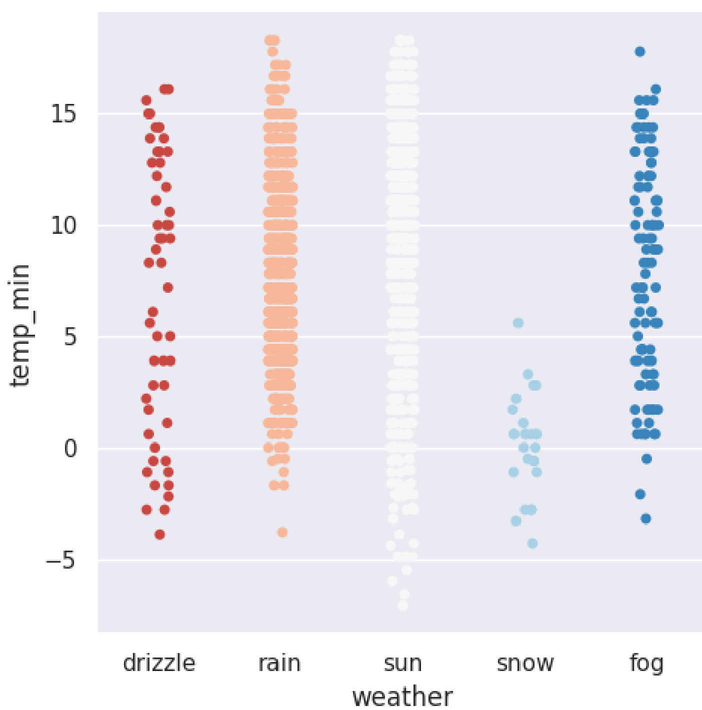

Figure 8 : Catplot for weather and minimum temperature

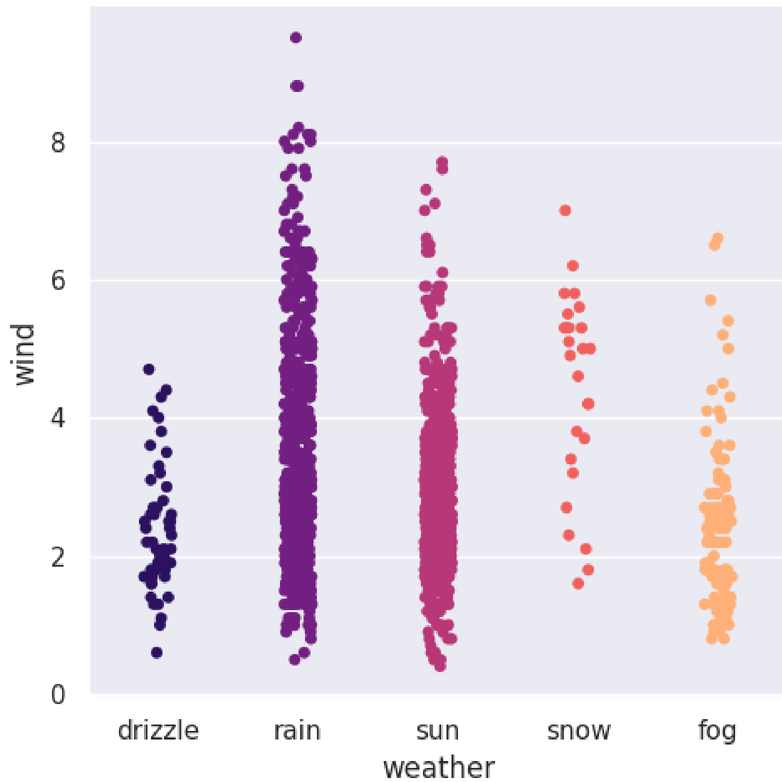

Figure 9 : Catplot for weather and wind

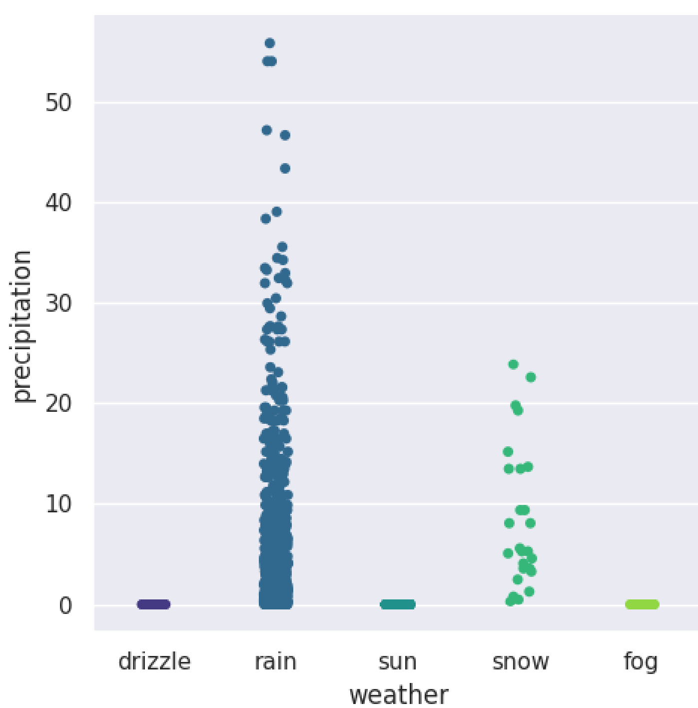

Figure 8 : Catplot for weather and precipitation

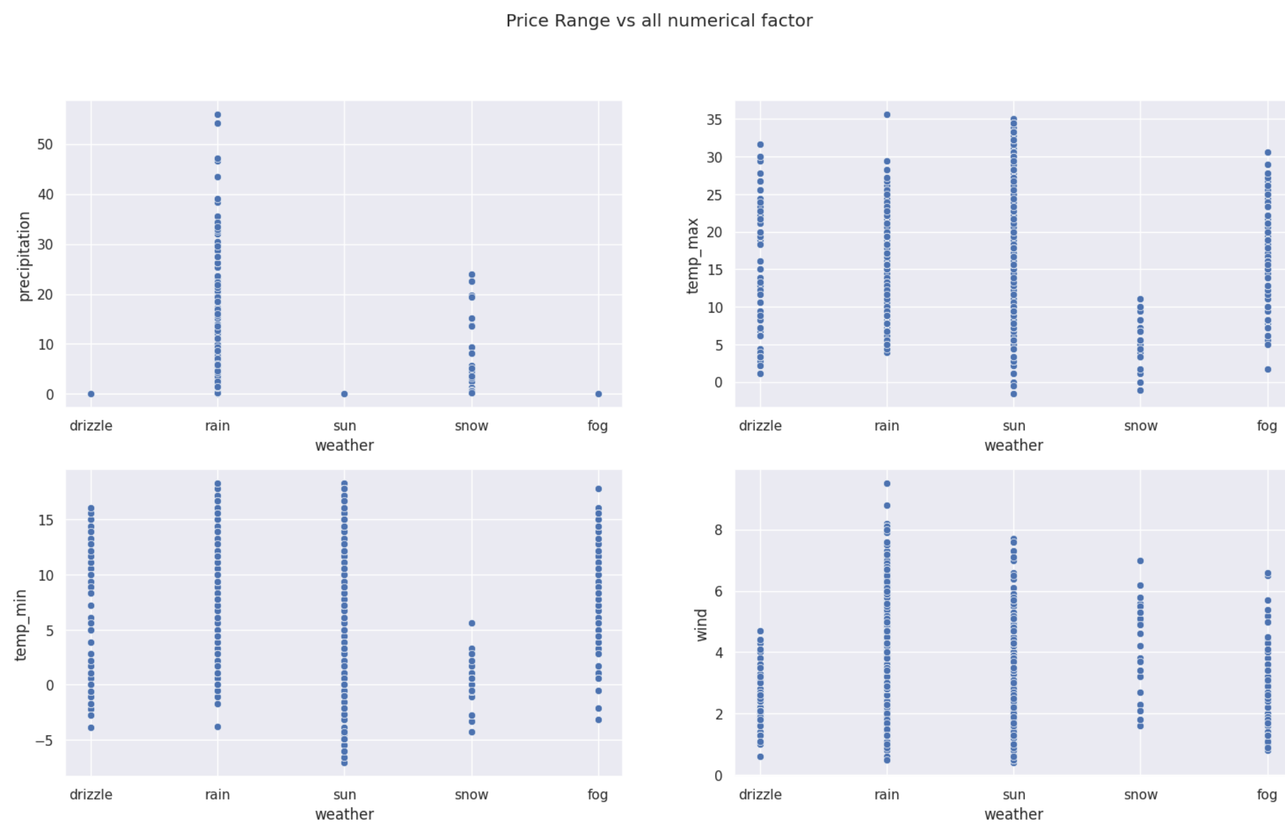

Figure 9 : Price range vs all numerical factor

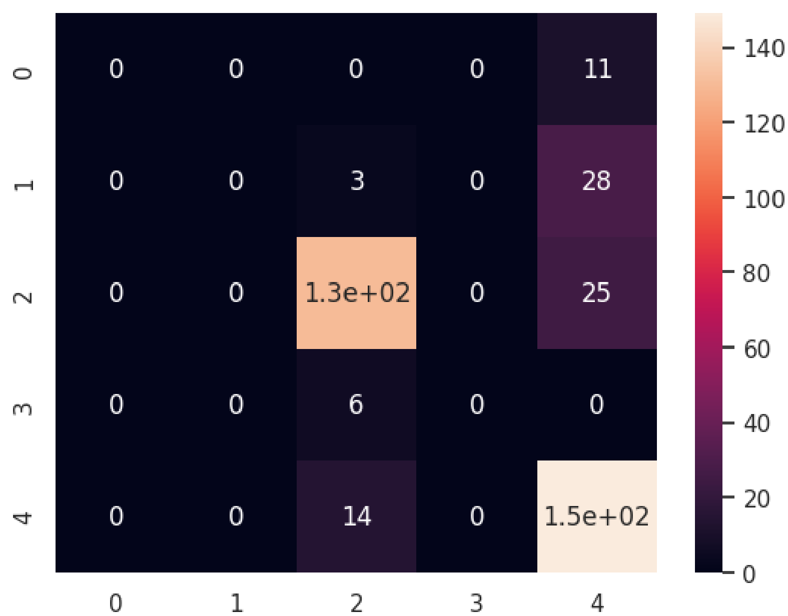

Figure 10 : Heatmap 1

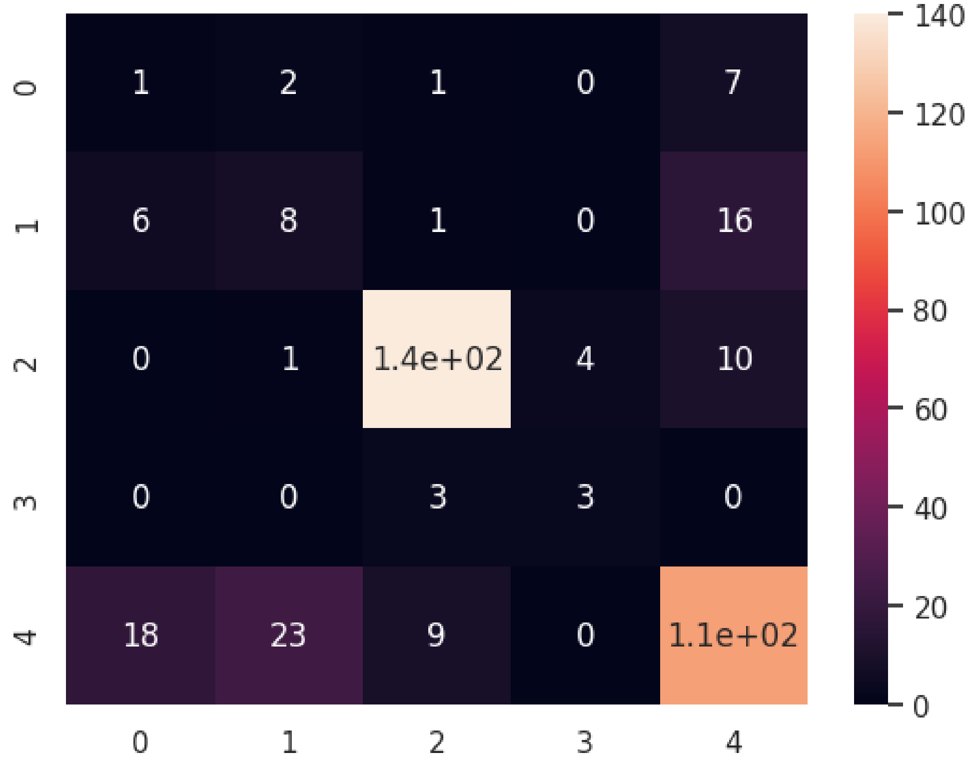

Figure 11 : Heatmap 2

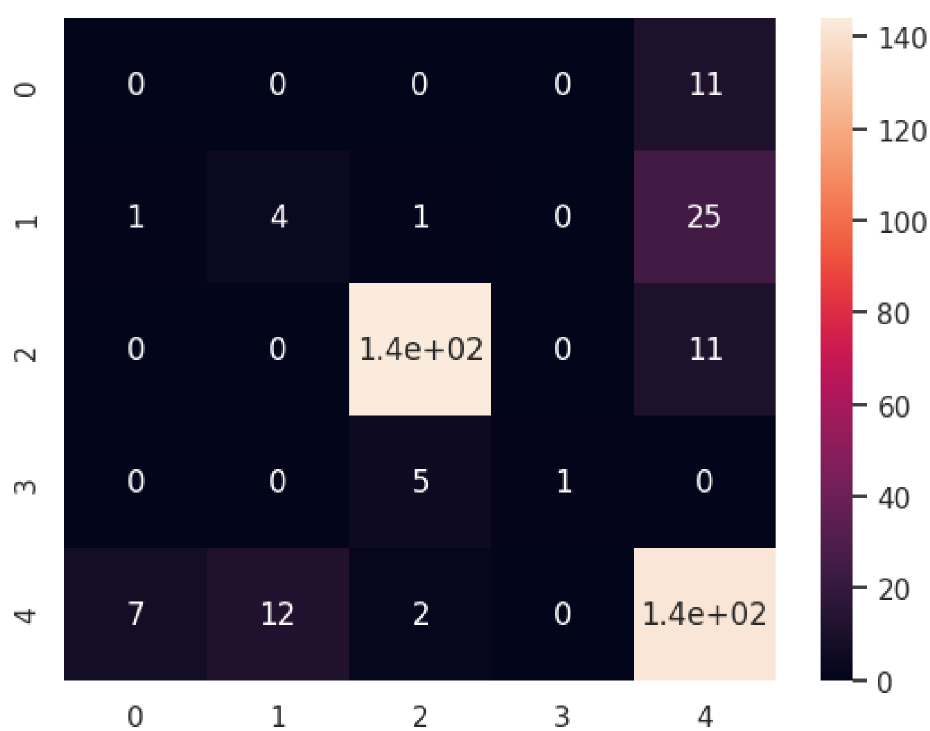

Figure 12 : Heatmap 3

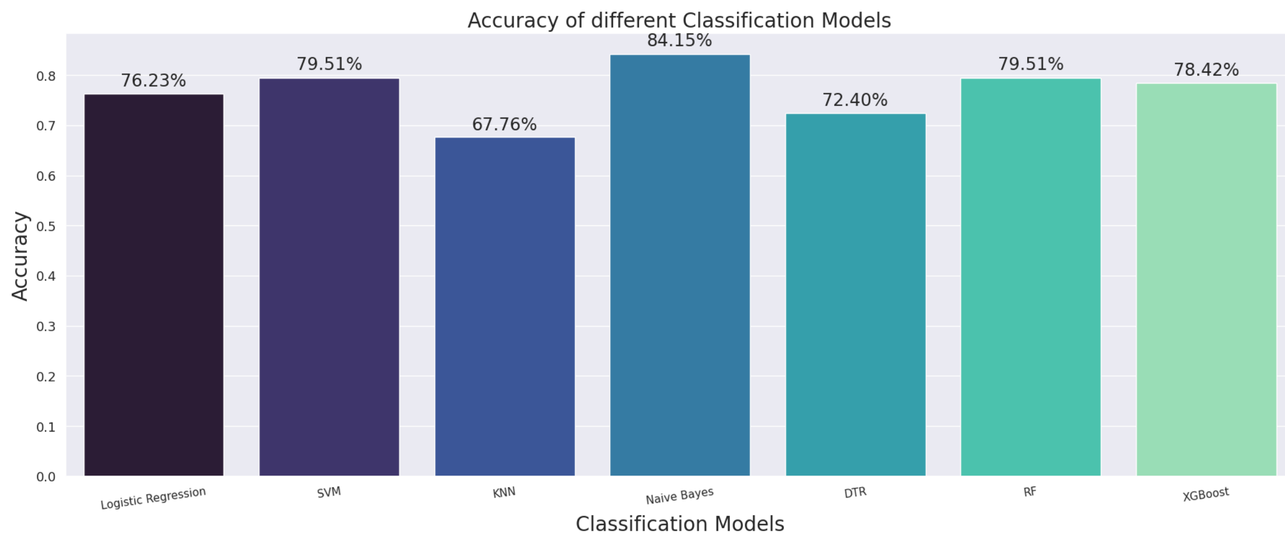

Figure 13 : Classification models and their accuracies

6.0 DEPLOYMENT

Naive Bayes is a straightforward but effective machine learning technique for classification applications. In the context of a weather app, it could be used to categorize weather conditions as sunny, cloudy, rainy, or snowy.

To build a weather app that uses a naive Bayes model classification, you must first gather a dataset of weather data. This information could include the date, time, location, temperature, humidity, wind speed, and precipitation. After you've gathered your data, you'll need to train a naïve Bayes model on it. This procedure entails feeding data to the model and allowing it to learn the correlations between the various features.

Once trained, the model can be used to classify new weather data. Simply feed the new data into the model, and it will return a prediction of the weather state.

There are several advantages of adopting a naive Bayes model categorization in a weather app. For starters, it is a rather straightforward method to implement. This implies it can be run on a wide range of platforms, including mobile devices, web servers, and cloud computing systems.

Second, naive Bayes models are extremely effective. This means that they may be used to classify vast amounts of data fast and efficiently. Finally, naive Bayes models are quite precise. This implies they can make accurate predictions about the weather.

So, in the future, we may use larger datasets to acquire more accurate results, but for now, Naive Bayes is the best option with the datasets of 1461.

7.0 DEMO VIDEO

https://youtu.be/bMyQRL4ydzA

8.0 CONCLUSION

In summary, the goal of this project was to create a model for weather prediction utilising historical weather data and machine learning techniques. We discovered insights about Seattle's weather patterns, variances, and trends from 2012 to 2015 by investigating and examining the "Seattle Weather" dataset. We studied the dynamics of temperature, precipitation, and wind speed using data visualisation and analysis and found relationships between meteorological variables.

Our research led us to design a machine learning algorithm-based weather prediction model. Future weather forecasts are produced by the model using historical weather data as input. We sought to increase the dependability and accuracy of weather forecasts by utilising data-driven methodologies, offering insightful information for decision-making across a range of industries.

There are significant consequences to the effective implementation of a trustworthy weather prediction model. It enables people and organisations to take well-informed decisions, allocate resources efficiently, and reduce risks brought on by bad weather. Accurate forecasts can help industries like agriculture plan their crops and schedule irrigation, while the transportation and tourism industries can streamline processes and reduce weather-related delays. Accurate weather forecasts also help emergency management organisations get ready for severe weather and increase societal resilience in general.

We have improved weather forecasting techniques throughout this project. We have shown the potential for enhancing forecasting precision and delivering trustworthy tools for decision-making processes by utilising historical weather data and machine learning algorithms.

The weather prediction model can be improved further in the future by adding new features and investigating more advanced machine learning techniques. The model will need to be evaluated and improved continuously in order to be reliable and effective in practical applications.

In conclusion, this project advances weather forecasting methods and lays the groundwork for additional study and invention in the future. We work to improve the accuracy of weather forecasts and their potential societal impact by leveraging the power of historical weather data and machine learning algorithms.

9.0 REFERENCES

1. Difference Between Encryption and Decryption. (2023, June 3). Guru99. https://www.guru99.com/difference-encryption-decryption.html
2. What is Sensitive Data? | UpGuard. (2022). Upguard.com. https://www.upguard.com/blog/sensitive-data
3. Robinson, P. (2019, November 19). *5 Benefits of Using Encryption Technology for Data Protection*. Lepide Blog: A Guide to IT Security, Compliance and IT Operations. https://www.lepide.com/blog/5-benefits-of-using-encryption-technology-for-data-protection/
4. Hussain, I., & Waghmare, C. (n.d.). *A Literature Review on Improvement of Weather Prediction By Using Machine Learning*. Retrieved June 16, 2023, from https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4140237
5. *Local Weather Forecast, News and Conditions*. (n.d.). Weather Underground. Retrieved June 16, 2023, from https://www.wunderground.com/
6. Bhaskar, A., & Hoque, N. (2021). Machine Learning-Based Weather Prediction Models: A Review. 2021 IEEE 8th International Conference on Industrial Engineering and Applications (ICIEA). doi: 10.1109/ICIEA52092.2021.9533301

-END OF DOCUMENT-